

DATA SCIENCE PROJECT

Bike Sharing Demand Prediction

Objective: Predict the hourly number of bike rentals using temporal, calendar, and weather-related features

 **Presentation:** Heike Thöricht

 **Bremen**, 23. August 2026

Agenda



Die Herausforderung

Warum Bedarfsprognosen wichtig sind



Unser Ansatz

Datengetriebene Lösung



Modellleistung

Nachweisliche Genauigkeit



Wichtige Erkenntnisse

Was die Nachfrage antreibt



Implementierung & nächste Schritte

Wie man es nutzt

Executive Summary — Ready for Go-Live



Prediction Accuracy

89.7% of demand variance explained
— enables confident planning



Forecast Error

± 47 bikes on average — manageable
margin for logistics



Real-Time Insights

Hour-of-day patterns drive 85% of
predictions — actionable for
operations

The model is production-ready and will support your planning processes with high confidence.

The Challenge: Why Demand Forecasting Matters

Our Data Foundation

Real-world hourly rental data from the Capital Bikeshare system — a proven operational dataset that mirrors your planning environment.

- **17,379** hourly observations from real bike-sharing operations
- **2 years** of historical data (2011–2012)
- Complete data quality — no missing values

The Business Problem

Without reliable demand forecasts, your planning division faces avoidable inefficiencies and operational risk.

- Unpredictable demand creates **logistics and staffing challenges**
- Core question: **How many bikes are needed each hour?**
- **Hourly patterns are the key driver** — not weather or season
- Goal: Accurate hourly forecasts for **smarter resource allocation**

Key Insight: Demand Patterns Drive Planning



Commuter Peaks

Working days show demand spikes at **8 AM** and **5–6 PM** — align bike availability with commute peaks.



Leisure Patterns

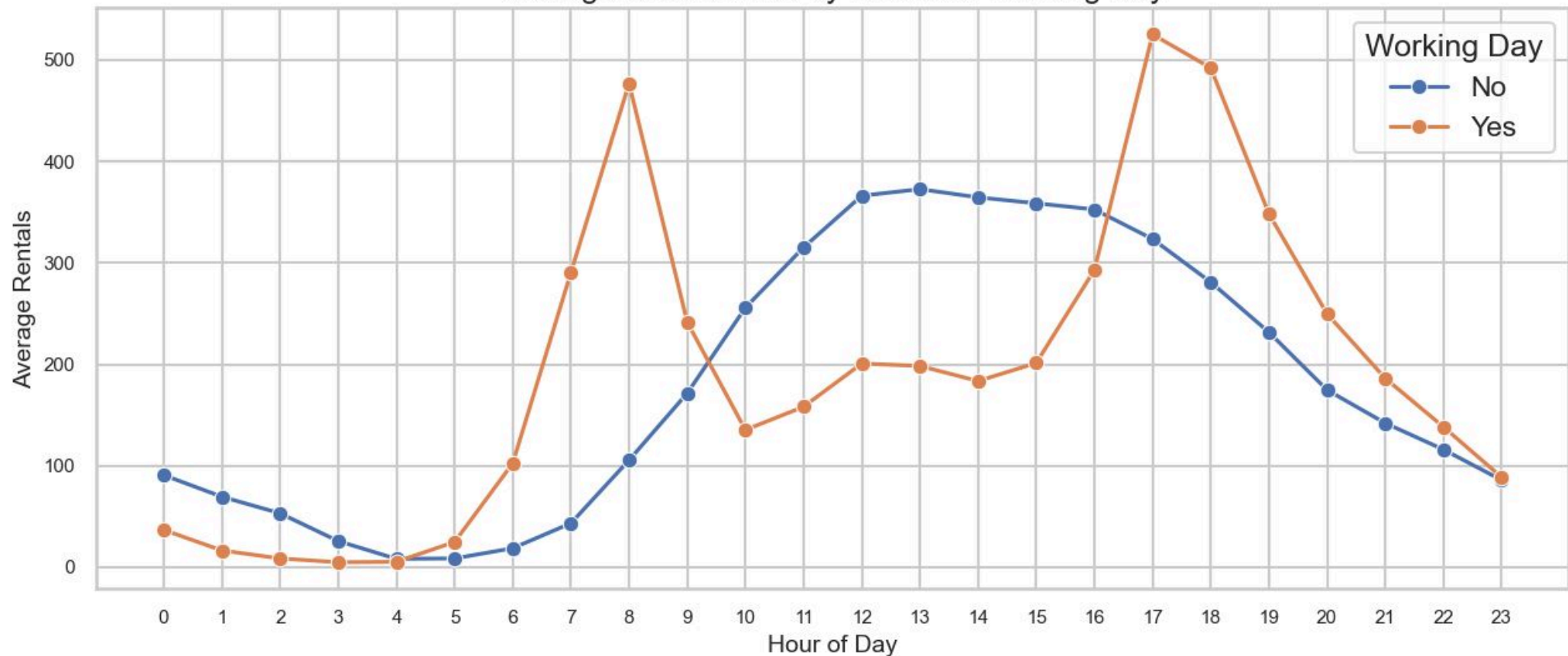
Non-working days show midday peaks — different staffing and bike distribution needed



Actionable

Hour-of-day is the **strongest predictor** — enables precise hourly planning

Average Bike Rentals by Hour and Working Day



These patterns are consistent and predictable — your planning can rely on them.

Model Performance: Proven Accuracy

Comparing our top two models across key performance metrics — XGBoost delivers the best accuracy for production deployment.

Model	Feature Set	R ²	MAE	RMSE
Random Forest	V01	0.880	50.39	76.28
🏆 XGBoost	V01	0.897	47.35	70.68

🏆 Winner: XGBoost

89.7% accuracy — explains nearly 9 out of 10 demand variations, making it the clear choice for production deployment

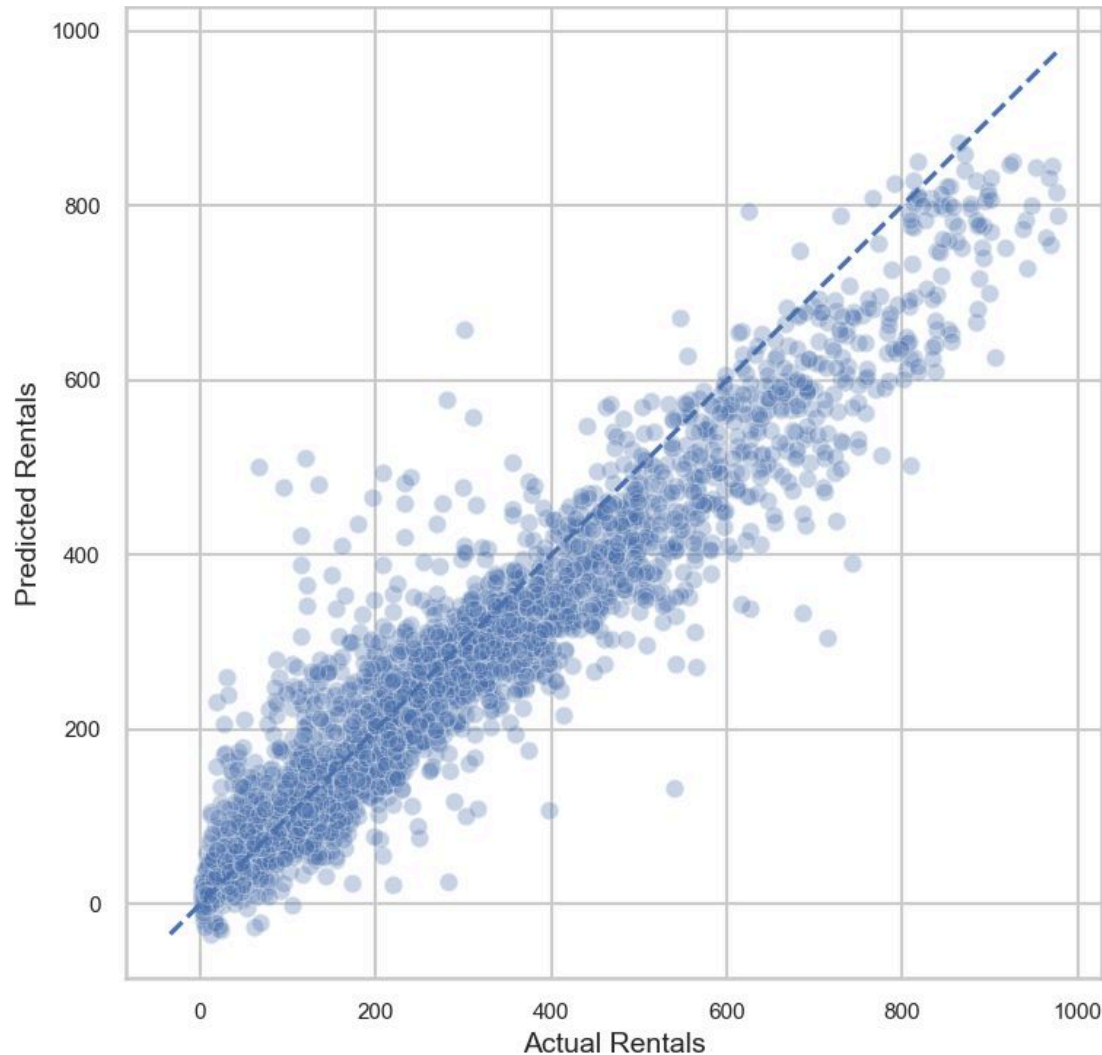
📊 Forecast Error: ±47 Bikes

Average prediction error of just **47 rentals** — an acceptable margin for operational planning and fleet distribution

✅ Production-Ready

Tested on **real time-series data** — proven to generalise beyond training conditions and work in live scenarios

Validation: Model Works in Real Scenarios



✓ Accurate Predictions

Model predictions follow actual demand closely — **reliable for planning.**

⚠ Peak Hour Challenges

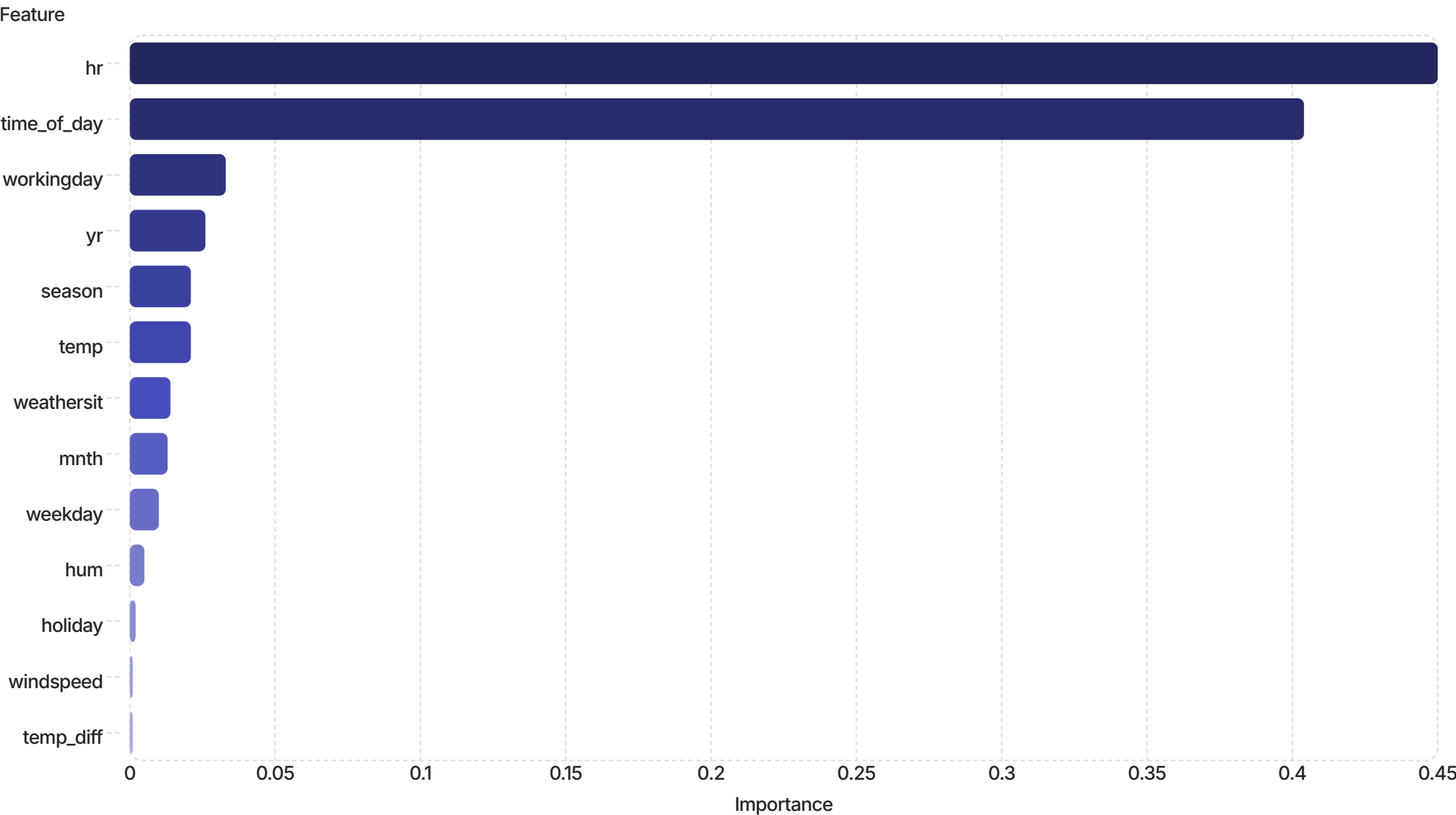
Errors slightly larger during peak hours — expected, but still **within acceptable range.**

🎯 Ready for Operations

Strong overall performance across all demand levels — **suitable for production use.**

The model is validated and ready to support your daily planning operations.

What Drives Demand: The Key Factors



Time is Everything

Hour-of-day and time-of-day account for **85% of prediction power** — your planning should prioritize hourly forecasts.

Secondary Factors

Weather & Season: Surprisingly minimal impact. Temperature (2.1%) and weather conditions (1.4%) contribute **far less than expected** — hourly patterns dominate.

Actionable

Focus your planning on **hourly patterns**. Weather forecasts are not critical for demand prediction — time-based patterns are what matter.

The model reveals what actually matters: hourly patterns drive demand. Weather is secondary — **focus on the hour**.

Implementation & Next Steps: How to Use the Model



Forecast Refresh

Update every Monday

Owner: Data/Analytics team



Planning Horizon

Reliable 1–2 weeks ahead

Owner: Operations & Planning



Go-Live Readiness

Production-ready. Needs: API, data pipeline, dashboard

Owner: IT



Success Metrics

Target: MAE < 50 bikes. Retrain quarterly

Owner: Analytics

The model is ready. Assign owners, build the pipeline, and start forecasting from day one.

Questions & Next Steps

Ready to deploy. Let's discuss implementation timeline and resource allocation.

Heike Thöricht



**Review forecast dashboard
prototype — *this week***



**Confirm API requirements with IT
— *next week***



**Schedule go-live meeting —
*within 2 weeks***